

THE AGENTOPIA TIMES: Understanding and Mitigating Hallucinations in Multi-Agent LLM Systems via Data Journalism Gameplay

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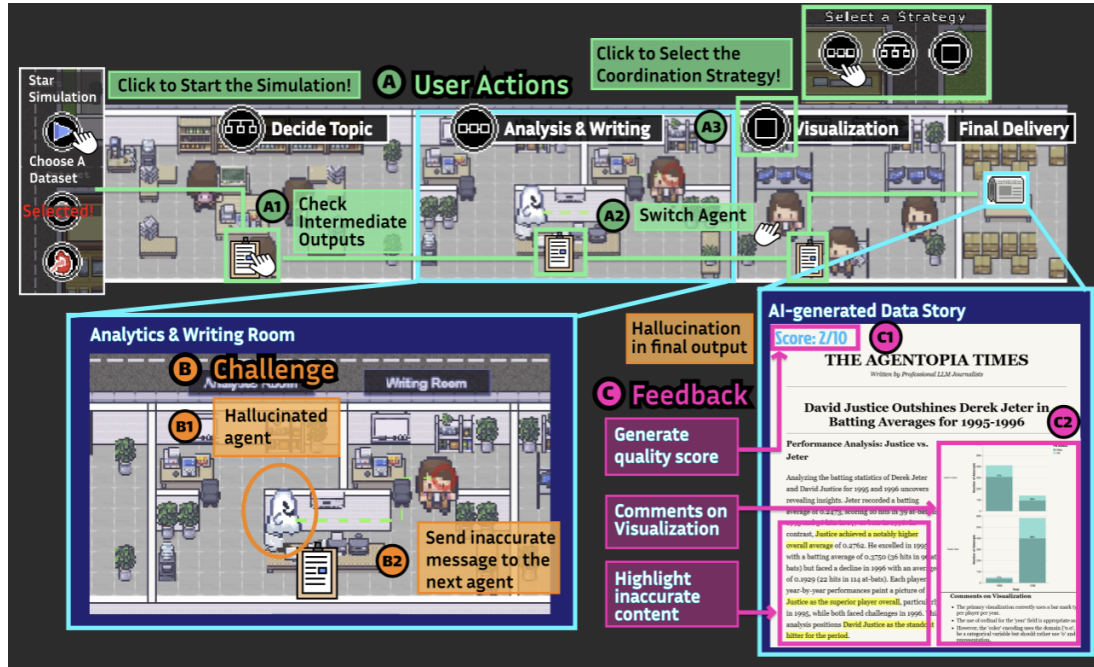


Figure 1: Overview of THE AGENTOPIA TIMES. This educational game is designed to teach hallucination mitigation in multi-agent system (MAS) with the integration of three key game components: (a) User Actions: enabling select datasets, run simulations, monitor intermediate agent outputs, and switch between agents; (b) Challenge: introducing hallucinated agents to simulate misinformation; (c) Feedback: providing quality scores, explanatory comments, and highlight hallucinated content. Players iteratively adjust their actions based on feedback to refine their strategies and address the game challenge.

ABSTRACT

Large language models (LLMs) are increasingly used to support data analysis and visualization tasks, but remain prone to hallucinations. Recent work suggests that multi-agent systems (MAS) can mitigate hallucinations by enabling internal validation and cross-verification. However, learning effective MAS coordination strategies to mitigate hallucination remains challenging, particularly for newcomers, due to the wide range of coordination strategies and the lack of interactive, hands-on learning tools. To address this, we present THE AGENTOPIA TIMES, an educational game that teaches hallucination mitigation through active experimentation with MAS coordination strategies. THE AGENTOPIA TIMES simulates a newsroom where LLM agents collaborate to create data-driven narratives, with users tasked with adjusting communication protocols to manage hallucinated content. The game features a structured mapping between MAS coordination and familiar gameplay mechanics, providing immediate feedback on hallucination outcomes. Through use cases

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and preliminary user feedback, we demonstrate how THE AGENTOPIA TIMES enables users to explore and mitigate hallucination in MAS.

Keywords: LLM, visualization generation, educational game, LLM hallucination, Multi-Agent.

1 INTRODUCTION

Large language models (LLMs) have recently gained widespread popularity and are widely used to assist users in data analysis and visualization creation, ranging from generating visualization code, evaluating design choices, and creating data-driven narratives [4, 22, 27]. However, LLMs often suffer from hallucinations, *i.e.*, generating content that appears plausible but is factually incorrect. For example, an LLM might fabricate nonexistent trends in a dataset, misinterpret statistical results, or produce misleading visualizations based on flawed reasoning [15].

Recently, multi-agent systems (MAS) have emerged as a promising approach to mitigate hallucinations and improve LLM performance by involving multiple agents with different roles and perspectives to work collaboratively. The coordination, redundancy, and internal validation among diverse agents can help cross-check outputs, detect inconsistencies, and improve overall reliability [5].

However, mitigating hallucination via efficient MAS architecture design can be challenging, especially for new AI practitioners and

students. This difficulty arises from the wide range of possible coordination strategies among agents, making it difficult for newcomers to compare approaches, understand trade-offs, and grasp the underlying principles of effective multi-agent collaboration. Previous studies [1, 26] have attempted to address this issue by offering high-level MAS design guidelines. While valuable, these resources often lack interactive, hands-on experiences that are essential for helping newcomers develop an intuitive understanding and bridge the gap between abstract principles and real-world application. Although some studies, such as the Smallville sandbox [13, 16], provide interactive environments for observing and evaluating agent behaviors, they provide limited support for users to modify or experiment with the underlying MAS architecture. This limits their utility in learning hallucination through trial and error.

To address these challenges, we draw inspiration from the proven effectiveness of games in education and propose an educational game, THE AGENTOPIA TIMES. As shown in Figure 1, THE AGENTOPIA TIMES models a newsroom where LLM agents collaborate to create data-driven stories, with a carefully-designed mapping from the MAS concepts to the familiar interactions in the game. Users are tasked to mitigate deliberately introduced hallucinations via experimenting with various agent communication protocols, gaining practical experience in understanding and mitigating hallucinations.

THE AGENTOPIA TIMES features a structured mapping between key MAS elements and familiar interactions in games, enabling intuitive learning through gameplay. Two use cases demonstrate how THE AGENTOPIA TIMES enables users to examine hallucination propagation and experiment with strategies to mitigate hallucination. The two main contributions of this work include: i) a novel game-based learning framework that transforms abstract multi-agent system concepts into interactive game experiences; ii) an educational game, THE AGENTOPIA TIMES, that is open-source and publicly accessible. We also reported preliminary user feedback of the game and discussed promising directions for future studies.

2 RELATED WORK

2.1 Interactive Systems for AI Education

Interactive systems for learning AI are increasingly popular with the wide adoption of AI. Previous studies have proposed many interaction visualization tools for understanding AI's internal working mechanism, such as CNN Explainer [25], Tensorflow Playground [19], and GNN 101 [14]. These tools enable users to examine and interact with the weights, neurons, and hidden layers inside a model via intuitive visual representations, making abstract AI concepts accessible. While such visualizations are valuable for AI education, they often lack support for a trial-and-error learning process, which is critical to build deep understanding via active experimentation and immediate feedback [?].

Game-based learning provides an effective approach for trial-and-error learning by giving users clear objectives that encourage active participation and feedback-driven learning. Studies showed that game-based learning significantly outperforms traditional methods in STEM education [9, 31]. For example, Explorable Explanations [2] support users in learning about various topics (e.g., math, physics, economics) by directly manipulating simulations of the underlying phenomena to achieve desired outcomes, such as adjusting the number of friends to manipulate trust dynamics in social science. However, most existing studies focus on general science rather than AI-related concepts. Most relevant to our study are Little Learning Machines [21] and Gamifying XAI [29]. The Little Learning Machines enables players to train reinforcement learning models, represented as miniature robots, to solve game challenges. While these systems demonstrate the power of game-based learning in AI, they cannot be directly applied to the unique challenges in MAS, which is the focus of this study.

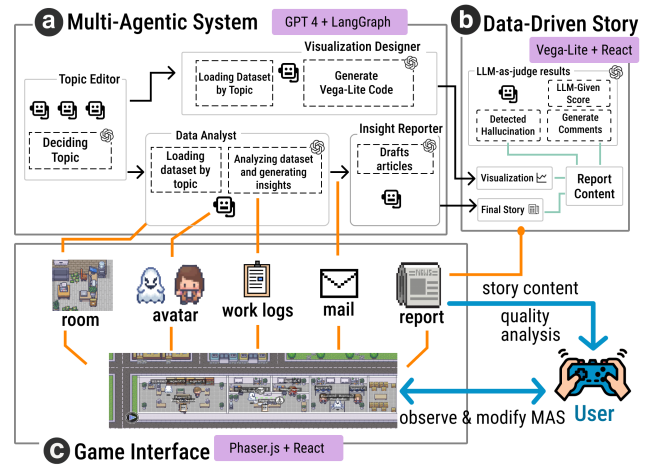


Figure 2: **System Overview of THE AGENTOPIA TIMES.** The system has three main components: (a) MAS Module, where agents collaborate to generate data-driven stories; (b) Data-Driven Story Module, which analyzes content and evaluates report quality using LLM-based feedback; and (c) Game Interface, where users interact with familiar game elements (e.g., avatars, logs, and reports) to observe and modify the MAS in a newsroom-themed game.

2.2 LLM for Visualization

Recently, there has been an increasing use of LLMs for visualization tasks, spanning areas such as visualization code generation, design recommendation, interaction simulation, and storytelling [6, 8, 10, 24]. For example, LIDA [8] introduces a multi-step, LLM-based pipeline for generating visualization and infographics for a given dataset. DataNarrative [10] proposes a multi-agent framework employing two LLM agents to generate data-driven storytelling.

As LLMs are increasingly adopted for visualization, concerns about hallucinations (i.e., the generation of incorrect or misleading information) have also emerged as a critical research topic. For example, ChartInsighter [23] summarizes the types of LLM hallucinations and mitigates them for time-series chart summaries. KnowNet [28] integrates knowledge graphs to mitigate LLM hallucination when generating node-link diagrams to illustrate health information. While recent efforts have focused on technical methods to detect and mitigate hallucinations, education for AI beginners that helps them understand and address hallucinations themselves remains underexplored. This gap forms the central focus of our study.

3 DESIGN GOALS

THE AGENTOPIA TIMES is guided by two primary design goals:

- G.1 Simplify the learning.** By mapping abstract MAS design patterns, such as role coordination and redundancy strategies, onto familiar and intuitive game interactions, the system aims to help non-expert users efficiently understand how the coordination among agents can help mitigate hallucination.
- G.2 Motivate engagement.** THE AGENTOPIA TIMES aims to promote active learning by giving users a clear game objective (i.e., mitigating hallucinations in the generated data-driven story). Timely feedback on user actions, including intermediate output, hallucination detection, and quality score of the report, fosters a trial-and-error learning process.

4 SYSTEM DESIGN AND IMPLEMENTATION

THE AGENTOPIA TIMES is an interactive educational game that allows users to examine and mitigate hallucinations in LLM agentic

systems. In the game, each agent is represented as a game character within a newsroom-themed environment, collaborating to analyze data and produce a data-driven story. Hallucinations are deliberately introduced into the system, allowing users to observe how they propagate through agent interactions. Users can experiment with different MAS collaboration patterns to mitigate hallucinations and improve the quality of the final report.

4.1 Game Interface

The game interface is designed to resemble a newsroom setting, featuring multiple office rooms and staff members actively at work.

As shown in Figure 2, the game interface (c) maps the key components of the Multi-Agent System (MAS) (a) as intuitive elements and interactions within a newsroom-themed game, with the aim of facilitating the learning experience of MAS (G.1).

Different office rooms correspond to distinct tasks involved in the MAS to produce a data-driven story, such as *Deciding Report Topics*, *Data Analysis*, *Visualization Generation*. In MAS, tasks are accomplished either by individual agents or through the collaboration of multiple agents. Reflecting this structure, the game interface represents each agent as a game avatar and assigns them to rooms for specific tasks. When multiple agents occupy the same room, they collaborate to complete the associated task. The collaboration pattern, such as agents independently completing the task and then voting, or one agent producing a result and another verifying it, is reflected in the spatial layout of the room.

All office rooms are arranged in a linear sequence from left to right, reflecting the task-executing order in MAS. The left-hand menu allows users to examine and select the dataset to be analyzed. Clicking the start button on the menu will pass the dataset to the leftmost room and start the process of generating a data-driven story. Once a task is completed, its output is passed to the next room, represented in the game environment as mail flying from one desk in one room to the next. A file icon is also attached to the room that sends the message, mimicking the work logs of the staff members and allowing users to click and review the intermediate output.

4.2 MAS Configuration

Each agent in THE AGENTOPIA TIMES is instantiated with a specific role and set of skills, which determine their behavior and output within the game. The system supports multiple types of agents, each fulfilling a distinct function in the data-to-story process:

- *Topic Editor*: Select the key themes to frame the story.
- *Data Analyst*: Process raw datasets to extract structured insights and patterns for story development.
- *Insight Reporter*: Draft initial article versions by weaving the structured insights into a coherent narrative.
- *Content Editor*: Refine drafts, improving structure and clarity.
- *Visualization Designer*: Work with data analysts to generate visualizations for the given dataset.

THE AGENTOPIA TIMES supports various collaboration patterns among agents, such as voting, sequential workflows, and routing mechanisms. For example, three *Topic Editor* agents independently generate candidate titles for the news article and then make a collective decision together, as shown in Figure 2(a).

The MAS operates on carefully selected datasets and produces data-driven storytelling. The inputs to the MAS are classic datasets that exhibit Simpson’s paradox, such as the Jeter vs. Justice dataset [17] and the Kidney Stone Treatment dataset [3]. Simpson’s paradox describes scenarios where a trend appears within individual groups of data but reverses or disappears when the groups are combined. These datasets were chosen because they are inherently misleading and present significant challenges for analysis. The MAS outputs include both textual content and executable Vega-Lite code, with the visualizations dynamically rendered and updated within the game interface, as shown in Figure 2(a).

4.3 Game Objectives and User Interactions

Game Objectives. Users of THE AGENTOPIA TIMES are tasked with a clear key game objective, *i.e.*, to mitigate hallucination in the generated data-driven story from the MAS simulation in the newsroom. The amount and type of hallucination are decided based on the current difficulty level of the game.

To maintain consistency and control over the gameplay experience, hallucination is deliberately introduced by injecting inaccurate statements about the dataset into the system prompts of selected agents. For example, “*Stone size in the dataset does not need to be analyzed*” will cause the agent to ignore important variables and generate incorrect output. This approach allows the system to precisely control both the amount and the type of hallucination.

To evaluate user performance on hallucination mitigation, an LLM-as-Judge evaluator assesses the factual accuracy of the generated data-driven story. The evaluator is provided with the ground-truth conclusion of the data analysis, and conducts rule-based scoring following predefined rubrics. These rubrics specify point deductions for common types of hallucinations and misinterpretations. For example, 5 points are deducted when outputs fail to identify the contradiction between subgroup trends and aggregate-level patterns, which is a key characteristic of Simpson’s Paradox. This evaluation approach aligns with recent studies on LLM-as-Judge [11, 30], which show that LLM-based evaluations achieve high accuracy and consistency when guided by clear instructions. We also clarified in the interface that certain coordination strategies (e.g., voting) tend to be more robust against hallucinations, and therefore are more likely to yield higher scores under the rubric-based evaluation.

User Interactions. To address the core objective of the game (*i.e.*, minimizing the negative impact of given hallucinations), users must carefully select i) which agent(s) to be placed into a hallucinating state and ii) how the hallucinating agent(s) collaborate with other agents, as these decisions are crucial for achieving a higher final score. Users can click on an agent to induce hallucination (visually indicated by a ghost-like avatar), and modify the coordination strategy within a room by clicking the strategy icon of the room, as shown in Figure 1A2-A3. In addition to the two key interactions, several supporting interactions are provided. Users can move to different areas of the newsroom via the WASD keys. Users can also select different datasets and start/pause the simulation in the left floating panel. Throughout the simulation, users can inspect both intermediate and final outputs by examining the work logs (*i.e.*, the file icon) associated with each room, as shown in Figure 2(c).

By providing a clear objective, familiar interactions to achieve this objective, and timely feedback, THE AGENTOPIA TIMES aims to create an engaging learning environment that effectively motivates user participation while reinforcing understanding of the hallucination dynamics (G.2).

4.4 Implementation

THE AGENTOPIA TIMES consists of two main components: a game interface and an MAS simulation engine. The **game interface** is built using Phaser.js [7], a lightweight HTML5 game framework. Avatars, rooms, and decorations are using the Modern Office tileset from LimeZu [12]. The generated data-driven report and the intermediate outputs of the MAS are displayed as pop-up windows within the game, implemented using React.js and Vega-Lite [18]. The **MAS simulation engine** is implemented using LangGraph.js [20], allowing easy agent orchestration. THE AGENTOPIA TIMES uses gpt-4o-mini for both the LLM agents and the LLM-as-Judge module. The source code of THE AGENTOPIA TIMES is available at <https://github.com/Visual-Intelligence-UMN/The-Agentopia-Times>, and an interactive demo is at <https://vis-llm-game.vercel.app/>.

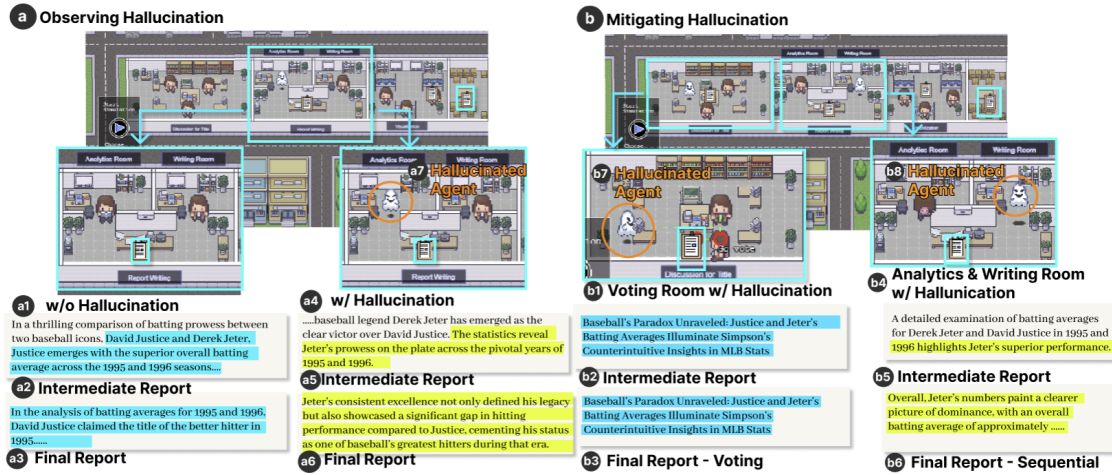


Figure 3: **Use Case.** a) Observing the propagation of hallucination in MAS; b) Mitigating hallucination via a voting collaboration. Yellow highlights: inaccurate information; blue highlights: accurate.

5 EVALUATION

5.1 Use Cases

We show how THE AGENTOPIA TIMES can help an LLM novice, Alex, understand and mitigate hallucination in MAS.

Observing Hallucination. Alex opens the game and selects the Justice and Jeter Baseball Players Dataset. In this dataset, Justice outperforms Jeter in each season, although Jeter has a higher overall average due to uneven at-bat distributions.

Alex compares simulations with and without hallucination. He first starts a simulation without hallucination. Once completed, Alex checks the final report, which scores 8/10 and contains no highlighted inaccuracies (Figure 3 a3). Then he clicks an agent in the Analytics Room and transforms it into a hallucinated agent, represented as a “ghost” (Figure 3 a7). After the simulation finishes, Alex clicks on the final report again to check the result. He finds there is a large area of highlighted inaccurate texts in the main paragraph, like “showcasing Jeter’s trajectory of ascension against Justice’s decline”, and the Scoring UI also only has 5/10 for now.

Mitigating Hallucination. To mitigate hallucinations, Alex experiments with assigning the hallucinated agent to different parts of the MAS as well as different collaboration patterns.

After dragging the ghost agent from the Analytics Room to the Writing Room (latter stage of the MAS), Alex clicked on the start button again to re-run the simulation. Alex checks the new report and finds that the quality has not improved, with the same 5/10 score. Alex looks into the intermediate output and finds some early signals of hallucinated information, such as “The results show that Jeter performs better than Justice...” (Figure 3 a5), propagated to the final report with highlighted texts such as “The results indicate that Jeter excelled in overall performance...”.

Thus, Alex decides to try to put that hallucinated agent into different workflow patterns, such as the voting pattern. When the third simulation is finished, Alex finds that the final quality score has a significant improvement from 5/10 to 8/10, with fewer hallucinated highlights (Figure 3 b2). On inspection, Alex finds that the Voting Room output was accurate and uninfluenced by the hallucinated agent (Figure 3 b7). This confirms his assumption that the voting mechanism mitigates hallucination by dampening individual bias.

5.2 Preliminary User Feedback

We conducted a preliminary user evaluation with five university students (P1-P5). All participants have a solid background in machine

learning, but limited knowledge of MAS. Each participant received a 5-minute tutorial, followed by free exploration of the system in their web browser using the think-aloud protocol, and ended the session with a feedback interview.

User Feedback & Observations. Overall, participants described the system as both enjoyable and educational. All users appreciated the system’s engaging interface, noting that “the characters are cute, and their behavior actually makes sense” (P3). Participants were able to develop hypotheses about effective MAS designs to mitigate hallucination and used the game interface to test these ideas. For example, P2 hypothesized that “voting might reduce hallucination” while P4 explored whether “hallucination happening at early stages might be fixed by later agents.”

At the same time, participants also provided valuable suggestions for improving usability, including in-situ hints about optional strategies for mitigating hallucinations, a clear indicator of the collaboration among agents and the intermediate hallucination propagation, and embedded tutorials on how to use the game interface.

6 CONCLUSIONS AND FUTURE WORK

In this paper, we present THE AGENTOPIA TIMES, an educational game that helps AI beginners understand and mitigate hallucinations in MAS. This work shows a new direction for interactive AI education, in which complex AI system behaviors are abstracted through intuitive game-based interactions, and learning is enhanced via structured and controllable challenges to the AI system.

For future work, we plan to expand THE AGENTOPIA TIMES by incorporating additional types of hallucinations, scaling up the evaluation, and improving the generality of the framework. First, we plan to introduce adjustable difficulty levels in the game by varying both types and quantities of hallucination. While the current implementation focuses on factual hallucinations (*i.e.*, agents output inaccurate information), we plan to explore more diverse types of hallucinations, such as omissions (*i.e.*, agents intentionally leave out critical information) and nonsense (*i.e.*, agents generate completely irrelevant responses to the input), to enrich the learning experience. Second, although preliminary user feedback was promising, larger-scale user studies with more diverse participants are needed to better evaluate both the user experience and the educational impact. Third, while the current game is set in a newsroom context, our underlying design framework applies to a wide range of contexts. In future work, we plan to integrate customizable APIs to enable flexible configuration of datasets, agent behaviors, and the visual tile sets of the game environment, therefore supporting broader use cases.

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